Machine Learning Model

Internship Task 2

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In [10]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

Load Dataset

In [11]:
```
import pandas as pd


url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv"
data = pd.read_csv(url)


data.head()
```

Out[11]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Feature Selection

Selecting input features and target variable

In [12]:
```
X = data[['total_bill']]
y = data['tip']
```

Train Test Split

Splitting dataset into training and testing data

In [13]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [14]:
```
print(X_train.shape)
print(X_test.shape)
```

```
(195, 1)
(49, 1)
```

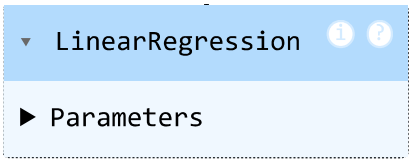
Model Training

Training the Linear Regression model

In [15]:
```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

Out[15]:



Prediction

Using the trained model to make predictions

In [16]:
```
predictions = model.predict(X_test)


predictions[:5]
```

Out[16]:
```
array([3.04525623, 1.86330727, 3.55119456, 3.69452593, 2.31576375])
```

Model Evaluation

Evaluating the performance of the model

In [17]:
```
from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)
```

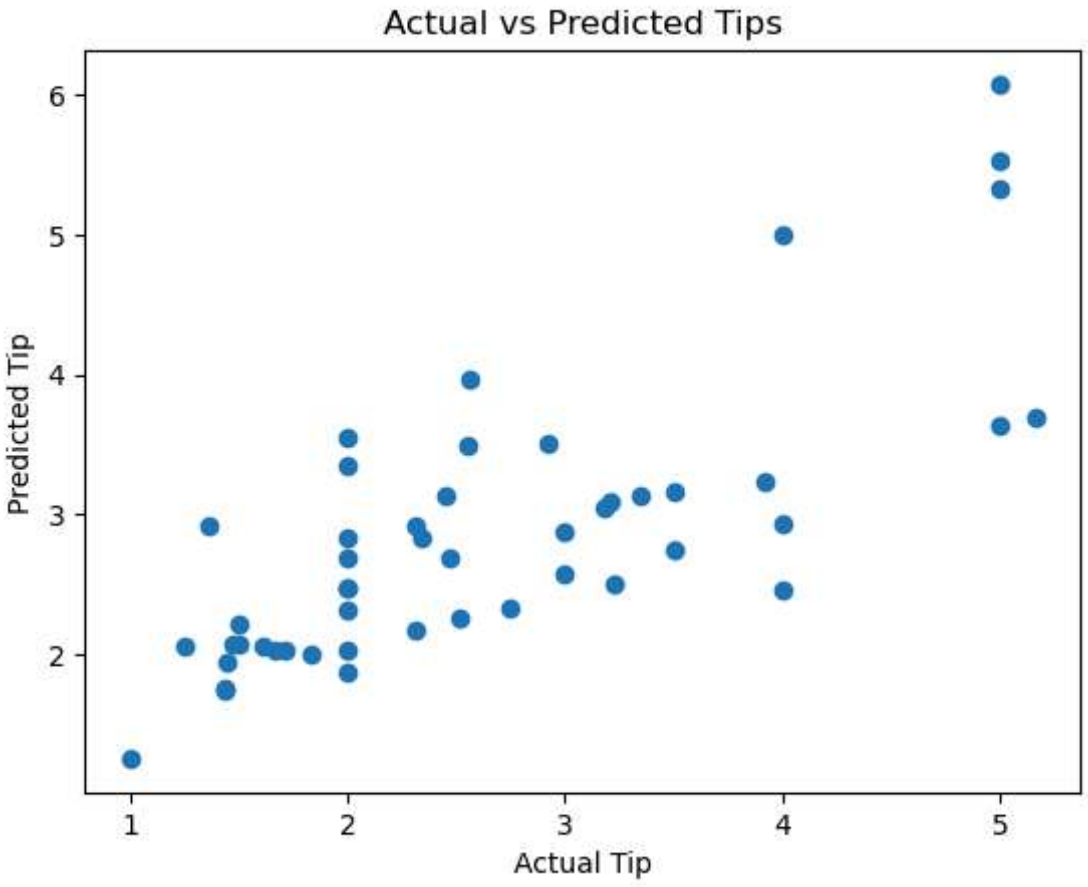
```
Mean Squared Error: 0.5688142529229538
R2 Score: 0.5449381659234664
```

Visualization

Comparing Actual vs Predicted values

In [18]:
```
import matplotlib.pyplot as plt

plt.scatter(y_test, predictions)
plt.xlabel("Actual Tip")
plt.ylabel("Predicted Tip")
plt.title("Actual vs Predicted Tips")
plt.show()
```



Conclusion

In this notebook, a Linear Regression model was built to predict tips based on the total bill amount. The dataset was split into training and testing sets. The model was evaluated

using Mean Squared Error and R2 Score, and the results were visualized using a scatter plot.

In [ ]: